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Pharmacodynamics of Piperacillin-Tazobactam/Amikacin Combination versus Meropenem against Extended-Spectrum β -Lactamase-Producing *Escherichia coli* in a Hollow Fiber Infection Model

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ABSTRACT Carbapenems are recommended for the treatment of urosepsis caused by extended-spectrum β -lactamase (ESBL)-producing, multidrug-resistant *Escherichia coli*; however, due to selection of carbapenem resistance, there is an increasing interest in alternative treatment regimens including the use of β -lactam-aminoglycoside combinations. We compared the pharmacodynamic activity of piperacillin-tazobactam and amikacin as mono and combination therapy versus meropenem monotherapy against extended-spectrum β -lactamase (ESBL)-producing, piperacillin-tazobactam resistant *E. coli* using a dynamic hollow fiber infection model (HFIM) over 7 days. Broth-microdilution was performed to determine the MIC of *E. coli* isolates. Whole genome sequencing was conducted. Four *E. coli* isolates were tested in HFIM with an initial inoculum of $\sim 10^7$ CFU/mL. Dosing regimens tested were piperacillin-tazobactam 4.5 g, 6-hourly, plus amikacin 30 mg/kg, 24-hourly, as combination therapy, and piperacillin-tazobactam 4.5 g, 6-hourly, amikacin 30 mg/kg, 24-hourly, and meropenem 1 g, 8-hourly, each as monotherapy. We observed that piperacillin-tazobactam and amikacin monotherapy demonstrated initial rapid bacterial killing but then led to amplification of resistant subpopulations. The piperacillin-tazobactam/amikacin combination and meropenem experiments both attained a rapid bacterial killing (~ 4 – 5 log₁₀) within 24 h and did not result in any emergence of resistant subpopulations. Genome sequencing demonstrated that all ESBL-producing *E. coli* clinical isolates carried multiple antibiotic resistance genes including *bla*_{CTX-M-15}, *bla*_{OXA-1}, *bla*_{EC}, *bla*_{TEM-1}, and *aac*(6')-Ib-cr. These results suggest that the combination of piperacillin-tazobactam/amikacin may have a potential role as a carbapenem-sparing regimen, which should be tested in future urosepsis clinical trials.

KEYWORDS extended-spectrum β -lactamase (ESBL), *Escherichia coli*, urosepsis, combination therapy, piperacillin-tazobactam and amikacin, dynamic hollow fiber infection model

Sepsis is a systemic, life-threatening medical emergency, caused by the dysregulated host response to an infection (1). Sepsis emerging from the urinary tract is known as urosepsis (2). Approximately 31% of all sepsis cases are urosepsis (3), and it is

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TABLE 1 Characteristics of ESBL-producing *E. coli* isolates tested in the SCK and HFIM experiments^a

<i>E. coli</i> isolates	MIC to TZP (mg/L)	MIC to MEM (mg/L)	MIC to AMK (mg/L)	Antibiotic resistance genes	MLST
CTAP#49	32	0.0625	8	<i>aac(3)-IIa, aac(6')Ib-cr, aadA5, bla_{CTX-M-15}, bla_{OXA-1}, catB3, dfrA17, mph(A), sul1</i>	131
CTAP#51	32	0.0625	8	<i>aac(3)-IIa, aac(6')Ib-cr, aadA5, bla_{CTX-M-15}, bla_{OXA-1}, catB3, dfrA17, mph(A), sul1</i>	131
CTAP#132	32	0.0312	8	<i>aac(3)-IIa, aadA5, aac(6')Ib-cr5, bla_{CTX-M-15}, bla_{EC}, bla_{OXA-1}, bla_{TEM-1}, acrF, emrD, mdtM, mph(A), catB3, sul1, tet(A), dfrA17</i>	410
CTAP#166	16	0.0625	8	<i>aac(6')Ib-cr5, bla_{CTX-M-15}, bla_{EC}, bla_{OXA-1}, acrF, emrD, mdtM, mph(A), catB3, tet(A)</i>	410

^aTZP, piperacillin-tazobactam; AMK, amikacin; MEM, meropenem; ESBL, extended-spectrum β -lactamase; MLST, multilocus sequence types; SCK, static-concentration time-kill; HFIM, hollow fiber infection model. European Committee on Antimicrobial Susceptibility Testing clinical breakpoint for *E. coli*: TZP, susceptible, ≤ 8 mg/L; resistant, > 8 mg/L; AMK, susceptible, ≤ 8 mg/L; resistant, > 8 mg/L; and MEM, susceptible, ≤ 2 mg/L; resistant, > 8 mg/L (46).

associated with significant morbidity and mortality in ICU patients (4). In urosepsis, *Escherichia coli* is the most prevalent pathogen (5), and the presence of extended-spectrum β -lactamase (ESBL) is common (6).

ESBLs are often encoded on plasmids, which also carry other genes encoding resistance to a wide range of antibiotics including non- β -lactams and β -lactam/ β -lactamase inhibitor combinations (7). The alarming rise of ESBL-producing *E. coli* with multidrug-resistant phenotypes (8) is associated with poor clinical outcomes, prolonged hospital stays, and elevated hospital-related costs (9). Rationally optimized antibiotic combinations have the potential to maximize bacterial killing and suppress the emergence of resistance (10). A particular antibiotic combination of escalating interest is that of a β -lactam/ β -lactamase inhibitor and an aminoglycoside, which is used in ICU for the treatment of infection caused by ESBL-producing *Enterobacteriales* (11).

Piperacillin-tazobactam is widely used in ICU for the treatment of urinary tract infections (UTIs) caused by ESBLs (12); it attains high urinary concentrations (13). However, ESBL-producing *E. coli* could be resistant to piperacillin-tazobactam if ESBLs are hyper produced (14) or via the coproduction of additional β -lactamases that are poorly inhibited by tazobactam (15). Resistance to piperacillin-tazobactam can also occur by different mechanisms including mutations in the active site of penicillin binding proteins, reduced porin expression, or elevated antibiotic efflux (16). In settings of increased resistance rates, monotherapy with piperacillin-tazobactam may be suboptimal. However, combination with aminoglycosides such as amikacin could broaden the spectrum of activity and suppress the emergence of resistance (17). Aminoglycosides are among the widely used antibiotics for UTIs caused by ESBLs, as they are mainly eliminated through the kidneys by glomerular filtration (18), and at least 97.6% of ESBL-producing *E. coli* remains susceptible to aminoglycosides such as amikacin (8).

In the MERINO study, piperacillin-tazobactam was not noninferior for 30-day mortality compared to meropenem for bloodstream infection (BSI) caused by ceftriaxone-nonsusceptible *E. coli* or *Klebsiella* spp. (19). However, the mortality difference for piperacillin-tazobactam was diminished after excluding patients randomized to piperacillin-tazobactam with isolates resistant by broth-microdilution (20). In urosepsis patients infected with piperacillin-tazobactam resistant ESBL-*E. coli*, piperacillin-tazobactam in combination with amikacin could be a good treatment option and a reasonable alternative to carbapenems.

As such, we compared antibacterial effects (bacterial killing and suppression of resistance emergence) of piperacillin-tazobactam and amikacin as mono and combination therapy versus meropenem monotherapy only, against four ESBL-producing, piperacillin-tazobactam resistant *E. coli* clinical isolates using a dynamic *in vitro* hollow fiber infection model (HFIM) over 7 days.

RESULTS

Phenotypic susceptibility, whole genome sequencing (WGS), and static-concentration time-kill experiments. Characteristics of *E. coli* isolates used in this study are presented in Table 1. In static-concentration time-kill (SCK), piperacillin-tazobactam in combination with amikacin at four different concentrations was synergistic at bacterial killing for each *E. coli* isolate (Fig. S1 in the supplemental material).

TABLE 2 Observed pharmacokinetic parameters during the first dosing interval against ESBL-*E. coli* isolates in the HFIM^a

<i>E. coli</i> isolates (MIC)	Dosing regimens	Observed			
		%fT _{>MIC}	fC _{max} /MIC	fC _{min} /MIC	fAUC/MIC
CTAP#49 (TZP, 32 mg/L; AMK, 8 mg/L; MEM, 0.0625 mg/L)	TZP 4.5 g, 6-hourly (TZP-AMK combination therapy)	56.4	5.8	0.3	
	AMK 30 mg/kg, 24-hourly (TZP-AMK combination therapy)		6.6	0.2	56.3
	TZP 4.5 g, 6-hourly (TZP monotherapy)	62.1	5.1	0.3	
	AMK 30 mg/kg, 24-hourly (AMK monotherapy)		6.3	0.3	57.0
	MEM 1 g, 8-hourly (MEM monotherapy)	100.0			
CTAP#51 (TZP, 32 mg/L; AMK, 8 mg/L; MEM, 0.0625 mg/L)	TZP 4.5 g, 6-hourly (TZP-AMK combination therapy)	53.8	5.6	0.3	
	AMK 30 mg/kg, 24-hourly (TZP-AMK combination therapy)		6.3	0.2	51.8
	TZP 4.5 g, 6-hourly (TZP monotherapy)	55.7	5.0	0.2	
	AMK 30 mg/kg, 24-hourly (AMK monotherapy)		6.4	0.2	54.3
	MEM 1 g, 8-hourly (MEM monotherapy)	100.0			
CTAP#132 (TZP, 32 mg/L; AMK, 8 mg/L; MEM, 0.0312 mg/L)	TZP 4.5 g, 6-hourly (TZP-AMK combination therapy)	62.4	5.1	0.3	
	AMK 30 mg/kg, 24-hourly (TZP-AMK combination therapy)		7.1	0.3	61.2
	TZP 4.5 g, 6-hourly (TZP monotherapy)	63.6	5.4	0.4	
	AMK 30 mg/kg, 24-hourly (AMK monotherapy)		7.2	0.3	51.1
	MEM 1 g, 8-hourly (MEM monotherapy)	100.0			
CTAP#166 (TZP, 16 mg/L; AMK, 8 mg/L; MEM, 0.0625 mg/L)	TZP 4.5 g, 6-hourly (TZP-AMK combination therapy)	77.6	10.2	0.5	
	AMK 30 mg/kg, 24-hourly (TZP-AMK combination therapy)		6.9	0.2	47.1
	TZP 4.5 g, 6-hourly (TZP monotherapy)	77.7	10.0	0.5	
	AMK 30 mg/kg, 24-hourly (AMK monotherapy)		5.9	0.2	46.6
	MEM 1 g, 8-hourly (MEM monotherapy)	100.0			

^aTZP, piperacillin-tazobactam; MEM, meropenem; AMK, amikacin; %fT_{>MIC}, the percentage of time that free concentrations exceeded the MIC of the pathogens; fC_{max}/MIC, free maximal concentration to the MIC; fC_{min}/MIC, free maximal concentration to the MIC; fAUC/MIC, the area under the free concentration-time curve to the MIC of the pathogens; HFIM, hollow fiber infection model.

HFIM experiments. (i) Pharmacokinetic profile. Simulated and observed pharmacokinetic parameters are presented in Table 2. The observed concentrations of piperacillin-tazobactam were close to the simulated concentrations where piperacillin-tazobactam was infused in combination with amikacin over 7 days (Fig. 1A, B). In HFIM, where piperacillin-tazobactam was infused as monotherapy, the concentrations gradually decreased and were substantially lower than the simulated values over 24–168 h (Fig. 1A, B; Table S3). Two presumed piperacillin degradation products, associated with ESBL hydrolysis, were observed. The concentrations of the presumed β -lactamase degradation products of piperacillin (termed as PPC1 and PPC2) estimated from their respective chromatogram peak area seem to account for the observed decline in piperacillin concentration (Fig. S2B, D). The simulated and observed concentration-time profiles for amikacin and meropenem are illustrated in Fig. 1C, D.

(ii) Effects of piperacillin-tazobactam/amikacin combination therapy and piperacillin-tazobactam or amikacin monotherapy against ESBL-*E. coli* isolates. Piperacillin-tazobactam in combination with amikacin resulted in ~ 4 – $5 \log_{10}$ CFU/mL of bacterial killing from the baseline within 24 h of treatment initiation (Fig. 2). Thereafter, the bacterial counts in this experimental arm were undetectable below the lower limit of quantification (LLOQ) within 28–34 h (Fig. 2). There was no observed emergence of resistant subpopulations over 168 h against any isolates (Fig. 2). Monotherapy with piperacillin-tazobactam attained initial bacterial killing of ~ 2 – $4 \log_{10}$ CFU/mL within the first 10 h, with extensive regrowth observed thereafter, surpassing the starting inoculum by 24–48 h. Subsequent doses did not attain any appreciable bacterial killing with bacterial regrowth paralleling the growth control for each *E. coli* isolate (Fig. 2). Simulated amikacin monotherapy reduced bacterial concentrations by ~ 1 – $2 \log_{10}$ and ~ 3 – $4 \log_{10}$ within the first 10 h against all isolates. Thereafter, a rapid regrowth, exceeding the initial inoculum, was observed against each isolate by 24 h (Fig. 2). Monotherapy with meropenem, on the other hand, resulted in $\geq 5 \log_{10}$ kill, decreasing the bacterial density to below the LLOQ within 24–28 h, and prevented regrowth over the treatment course (Fig. 2 and 3).

(iii) Effects of combination therapy and monotherapies on emergence of resistance. No detectable resistant subpopulations were observed from the piperacillin-tazobactam and amikacin combination arm against any isolates at 168 h (Fig. 3). For piperacillin-tazobactam

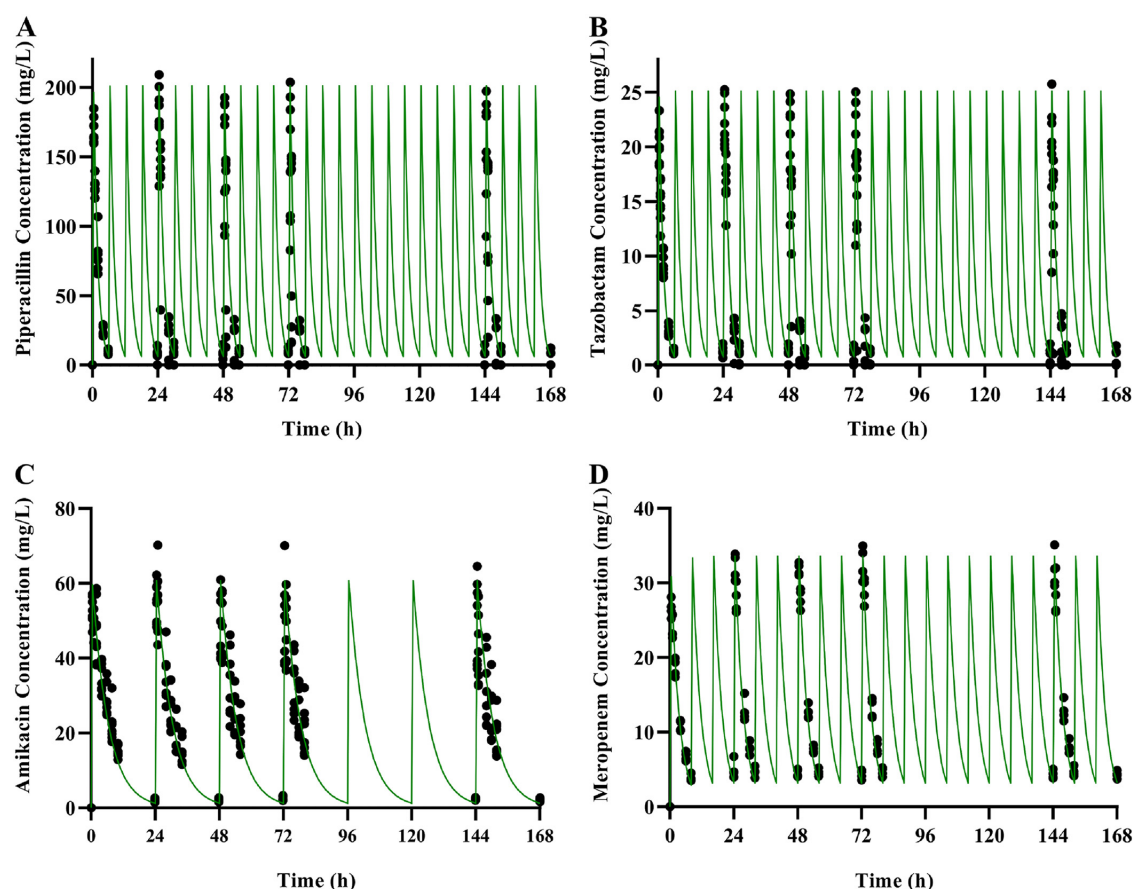


FIG 1 Pharmacokinetic profiles for the HFIM experiments with ESBL-producing *E. coli* isolates. Simulated (solid green curves) and observed (filled black circles) concentration-time profiles for (A) piperacillin, (B) tazobactam, (C) amikacin, and (D) meropenem. HFIM, hollow fiber infection model.

and amikacin monotherapy arms, there was a regrowth of resistant subpopulations on piperacillin-tazobactam or amikacin containing Cation-Adjusted Mueller-Hinton Agar (CAMHA) (Fig. 3). The piperacillin MICs of bacterial population that grew on antibiotic-free plates were 8-fold higher than the baseline MIC of 32 mg/L for CTAP#51, 16-fold higher than the baseline MIC of 32 mg/L for each of CTAP#49 and CTAP#132, and 32-fold higher than the baseline MIC of 16 mg/L for CTAP#166 *E. coli* isolate at 168 h with the piperacillin-tazobactam-treated arm. The amikacin MICs of bacterial population that grew on antibiotic-free CAMHA were 16-fold higher for CTAP#132, 32-fold higher for CTAP#49 and CTAP#51, and 64-fold higher for CTAP#166 than the baseline MIC of 8 mg/L for each *E. coli* isolate at 168 h with amikacin-treated arm (Table S2).

The piperacillin MICs of the bacterial population that grew on antibiotic-free CAMHA with growth control was 4-fold higher than the baseline MIC of 32 mg/L for CTAP#51, 8-fold higher than the baseline MIC of 32 mg/L and 16 mg/L for CTAP#49 and CTAP#166, respectively, and 16-fold higher than the baseline MIC of 32 mg/L for CTAP#132 at 168 h (Table S2). The amikacin MICs of bacterial population that grew on antibiotic-free CAMHA with growth control at 168 h were 8-fold higher for CTAP#166, 16-fold higher for CTAP#51 and CTAP#132, and 32-fold higher for CTAP#49 than amikacin baseline MIC of 8 mg/L for each *E. coli* isolate (Table S2).

DISCUSSION

To the best of our knowledge, this is the first HFIM study comparing the pharmacodynamic activity of piperacillin-tazobactam and amikacin as mono and combination therapy versus meropenem as monotherapy only against ESBL-producing, piperacillin-

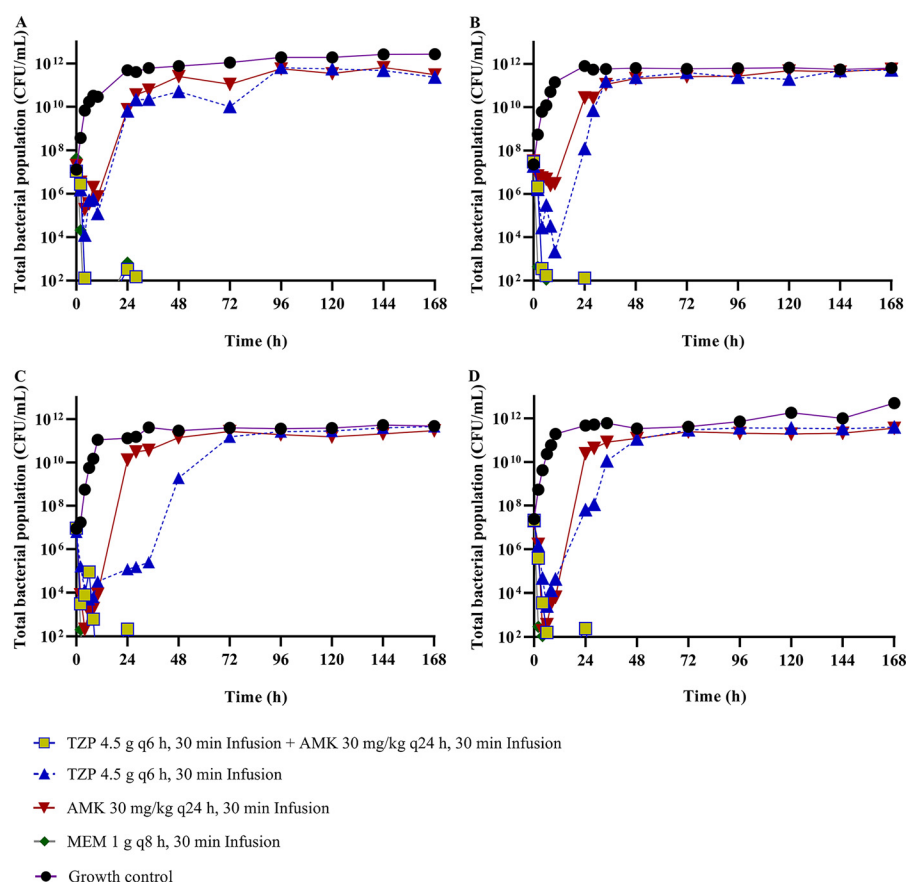


FIG 2 Effects of piperacillin-tazobactam and amikacin as mono and combination therapy on the concentration of the total bacterial population of ESBL-producing *E. coli* (A) CTAP#49, (B) CTAP#51, (C) CTAP#132, and (D) CTAP#166 versus the growth with meropenem as monotherapy in the HFIM. TZP, piperacillin-tazobactam; AMK, amikacin; MEM, meropenem; q6 h, 6-hourly; q8 h, 8-hourly; q24 h, 24-hourly; HFIM, hollow fiber infection model.

tazobactam resistant *E. coli* clinical isolates. This study has two key findings. Firstly, the bactericidal activity of the combination of piperacillin-tazobactam and amikacin was comparable to that of meropenem. Secondly, the combination regimen suppressed regrowth of resistant subpopulations, unlike monotherapy with either drug. Monotherapy with meropenem also suppressed regrowth of resistant subpopulations.

The potentiation of β -lactam antibiotics with aminoglycosides has been described mostly for *Pseudomonas aeruginosa*. Our results demonstrate that piperacillin-tazobactam and amikacin combination attained ~ 4 – $5 \log_{10}$ bacterial killing within 24 h and prevented emergence of resistance, which is in agreement with a previous HFIM study for *P. aeruginosa* (21, 22). The synergistic pharmacodynamic interaction observed in our study may be partly due to β -lactam-mediated bacterial cell wall disturbance, which in turn facilitates the uptake of aminoglycosides (23). The other mechanism could be aminoglycoside-mediated disruption of bacterial outer membrane, which could enhance the target site penetration of piperacillin (24). For example, Yadav et al. (25) demonstrated that *P. aeruginosa* outer membrane disruption by tobramycin enhances target site concentrations of imipenem. Therefore, synergistic combination of piperacillin-tazobactam with amikacin, coupled with tazobactam-mediated ESBL inhibition, may be partly attributable to the observed bacterial killing of ~ 4 – $5 \log_{10}$ CFU/mL in our study (Fig. 2), where the immune system could eradicate any residual bacteria (26).

Tazobactam is a suicidal β -lactamase inhibitor, which helps to safeguard the efficacy of piperacillin in the presence of most ESBL variants such as TEM, SHV, and CTX-M β -lactamases (27). However, for all ESBL-producing *E. coli* isolates in our study,

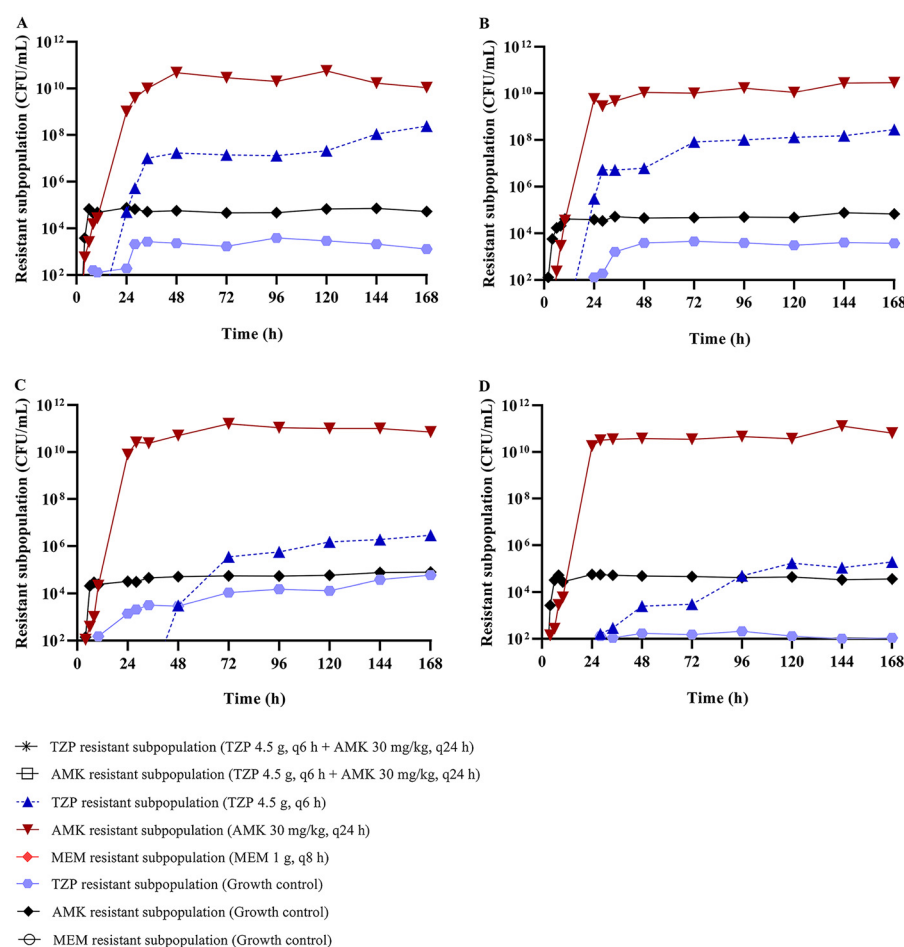


FIG 3 Effects of piperacillin-tazobactam and amikacin as mono and combination therapy on the density of resistant subpopulation of ESBL-producing *E. coli* (A) CTAP#49, (B) CTAP#51, (C) CTAP#132, and (D) CTAP#166 with increased MIC from baseline in the HFIM. CAMHA, cation-adjusted Mueller-Hinton agar; TZP, piperacillin-tazobactam; AMK, amikacin; MEM, meropenem; q6 h, 6-hourly; q8 h, 8-hourly; q24 h, 24-hourly; HFIM, hollow fiber infection model.

monotherapy with piperacillin-tazobactam resulted in rapid regrowth of piperacillin-tazobactam-resistant subpopulations, except for CTAP#132, where piperacillin-tazobactam-resistant subpopulations surpassed that of growth control only after 72 h. These findings are consistent with previous findings of its utility against *P. aeruginosa* (21). We observed carriage of multiple β -lactamase encoding genes such as *bla*_{CTX-M-15}/*bla*_{OXA-1}/*bla*_{EC}/*bla*_{TEM-1} in ESBL-producing *E. coli* clinical isolates tested in the present study (Table 1). For the piperacillin-tazobactam monotherapy arm, the observed concentrations of the drugs progressively decreased from the simulated concentrations over the treatment course (Fig. 1A and B, Table S3), and two presumed piperacillin degradation products were observed (Fig. S2B, D). Our observations are consistent with a previous HFIM study, where they observed the degradation of ceftaroline in the absence of avibactam against β -lactamase-producing *Enterobacteriales* (28). In a later study, degradation of ceftazidime was also reported when the agent was administered as monotherapy (29). This degradation may be attributable to the overexpression of ESBLs (27), carriage of multiple β -lactamases (15), or mutations in β -lactamase encoding genes (30). Amplification of ESBL genes upon exposure to antibiotics that leads to ESBL enzyme hyperproduction was described as the primary mechanism for β -lactam/ β -lactamase inhibitor resistance in *E. coli* isolate (31). These *in vitro* data observed that ESBL *bla*_{TEM-1} levels were 3.7-fold higher in the presence of piperacillin-tazobactam versus no antibiotic (31). WGS revealed that in *E. coli* the *bla*_{TEM-1} gene is located in a 10-kb

genomic resistance module (GRM) region that harbors multiple resistance genes and mobile genetic elements. The GRM was seen to be tandemly repeated at least 5-fold within a p1ESCUM/p1ECUMN-like plasmid when bacteria were grown in the presence of piperacillin-tazobactam. For example, in the presence of 8/8 mg/L piperacillin-tazobactam, the GRM copy number per cell increased approximately 10-fold to >90, which reflects at least an ~800-kb increase in genomic DNA and contribution of greater than 15% to the antibiotic resistance during piperacillin-tazobactam treatment (31). In the present study, the piperacillin-tazobactam 4.5 g 6-hourly regimen seemed to have insufficient tazobactam for the amount of ESBL present. The tazobactam was therefore consumed in the enzyme-substrate interaction, leading to subsequent piperacillin degradation by the presence of hyper-produced ESBL.

The inhibitor activity depends on the turnover number (t_n), which is described as the number of inhibitor molecules that are hydrolyzed per unit time before an enzyme molecule is completely inactivated, and these interactions are unique for each type of β -lactamase (32). For example, tazobactam appeared to inactivate the PC1 penicillinase ($t_n = 2$) more rapidly than the TEM-2 β -lactamase ($t_n = 125$) (27). Thus, tazobactam may interact differently against multiple β -lactamase encoding genes such as *bla*_{CTX-M-15/OXA-1/EC/TEM-1} harboring *E. coli* isolates in our study. Additionally, *E. coli* isolates expressing *bla*_{CTX-M-15/OXA-1} genotypes are associated with elevated piperacillin-tazobactam MIC compared to those expressing only *bla*_{CTX-M-15} (20). Thus, treatment failure with piperacillin-tazobactam monotherapy in the current study may be attributable to carriage and overexpression of multiple β -lactamases, particularly carriage of *bla*_{CTX-M-15/OXA-1/EC/TEM-1} in *E. coli* clinical isolates (presented in Table 1). Additionally, mutations within the amplified genes may allow cells to develop more durable resistance to antibiotics (31). For example, *E. coli* carriage of *bla*_{TEM-1} ESBL gene on a multicopy (19 copies per cell) plasmid was tested with increasing concentrations of ceftazidime (33). Mutations in *bla*_{TEM-1} were proliferated by increased expression produced from a high copy number of plasmids as well as due to the increased *bla*_{TEM-1} gene dosage (33), thereby conferring resistance to ceftazidime with an increasing rate of ceftazidime hydrolysis (33). Furthermore, transposons could promote the evolution of resistance islands in *E. coli* isolates. Genes associated with transposable elements generally showed higher levels of expression upon antibiotic-treated conditions compared with antibiotic-free conditions against multidrug-resistant *Acinetobacter baumannii* (34). These transposable elements demonstrated a higher probability of being cotranscribed with drug inactivation enzymes, and their associated antibiotic resistance genes were more likely to be upregulated in multidrug-resistant strains. For instance, transposon-associated *mel* and *mph* were cotranscribed, and both contributed to amikacin resistance (34).

Unlike monotherapy with piperacillin-tazobactam in this HFIM, no resistance proliferation and no loss of piperacillin concentration were observed in the treatment arm where piperacillin-tazobactam was dosed in combination with amikacin (Fig. 3 and Fig. S2A, C). The addition of amikacin to piperacillin-tazobactam resulted in synergistic bacterial killing and prevented emergence of resistance. No potential degradation of piperacillin by overexpression of ESBLs or multiple other β -lactamases was observed in combination arm (Fig. S2A, C), which is likely attributable to the rapid bacterial killing (Fig. 2) obtained early after combination therapy commencement.

For amikacin monotherapy, on the other hand, an initial bacterial kill was observed within the first 10 h, followed by a rapid regrowth of amikacin-resistant subpopulations within 24 h. These resistance amplifications may be in part due to the presence of the *aac(6)-Ib-cr* gene (35) in *E. coli* clinical isolates used in the present HFIM (Table 1). The *aac(6)-Ib-cr* gene is a variant of the *aac(6)-Ib* gene coding for aminoglycoside acetyltransferase that mediates acetylation and is responsible for resistance to aminoglycosides (36).

Clinical studies investigating the efficacy of the piperacillin-tazobactam/amikacin combination in urosepsis caused by ESBL-producing *E. coli* are limited. In a meta-analysis, β -lactam/single daily dose aminoglycoside combination was not significantly associated with reduced mortality ($n = 3,686$, OR 0.82, 95% CI 0.63–1.08, $P = 0.10$, P^2 42%) when compared with β -lactam monotherapy in patients with sepsis or septic shock (37). However,

the results between treatment arms may be attributable to heterogeneous patient populations, different aminoglycosides used with β -lactams, and smaller sample size of the included studies (37). Therefore, clinical trials with robust design investigating this particular context are required.

This study has limitations. Firstly, *in vitro* HFIM lacks the host immune response, in the presence of which the exposure-response relationship may be different, as observed in clinical practice. Nonetheless, the HFIM is able to forecast clinical microbiological outcomes with high accuracy (38). Secondly, we have used four clinical *E. coli* isolates; therefore, results from this study could be interpreted with caution for other members of *Enterobacterales*. Thirdly, because of the high expense of running HFIM, we opted to test multiple isolates and instated duplicates of just two isolates, which is a common approach, as described for other HFIM studies (39, 40). Fourthly, only one renal function (CrCL of 100 mL/min) was used. However, we have used an approximate median value that is representative of the target population (41).

In conclusion, piperacillin-tazobactam and amikacin combination therapy in our study was comparable to meropenem monotherapy in bacterial killing and suppression of emergence of resistance against all ESBL-producing, piperacillin-tazobactam resistant *E. coli* clinical isolates. Given the practical challenges of using the gold-standard broth-microdilution MIC assay in routine clinical practice, the use of an aminoglycoside in combination with piperacillin-tazobactam in the initial phase of treatment for ESBL producers could be a useful option until accurate susceptibility data are available for the isolated pathogen. Future studies should be performed with bacterial isolates with varying piperacillin-tazobactam and amikacin MICs, which could help to characterize the use of an aminoglycoside in combination with piperacillin-tazobactam in a much wider context of bacterial isolates with different MIC values. Further evaluation in randomized controlled trials is also required to support our results.

MATERIALS AND METHODS

Microorganisms and antibiotics. Four ESBL-producing *E. coli* clinical isolates were used in this study. CTAP#49 and CTAP#51 were collected from the MERINO randomized controlled trial (42), *E. coli* CTAP#132 was obtained from Zowawi et al. (43), and *E. coli* CTAP#166 was obtained from the national reference laboratory for antimicrobial resistance surveillance in Bangladesh (44). Bacterial isolates were stored in Cation-Adjusted Mueller-Hinton Broth (CAMHB) (BD, Sparks, MD, USA) containing 20% glycerol vol/vol at -80°C .

Piperacillin analytical standards (Sigma-Aldrich, lot 098M4886V), tazobactam (Fluorochem, lot FCB009363), amikacin (Fluorochem, lot FCC1886), and meropenem (Tokyo Chemical Industry Co., Ltd., lot YCY8L-BF) powder were used to prepare stocks using Milli-Q water, filter sterilized using $0.22\text{ }\mu\text{m}$ polyvinylidene fluoride syringe filter and stored immediately at -80°C . These stocks were used for *in vitro* susceptibility testing, SCK assays, and preparing piperacillin-tazobactam, amikacin, or meropenem containing Cation-Adjusted Mueller-Hinton Agar (CAMHA, BD, Sparks, MD, USA) plates. For HFIM dosing, stock solutions were prepared from clinical intravenous formulations of piperacillin-tazobactam (PipTaz, 4 g/0.5 g, AFT pharmaceuticals, Australia), amikacin (Amikacin sxp 500 mg/2 mL, Southern Cross Pharma, Australia), and meropenem (Meropenem, 1,000 mg, Fresenius Kabi, Australia) and immediately stored at -80°C .

Phenotypic susceptibility testing. The MICs of piperacillin-tazobactam, amikacin, and meropenem for *E. coli* clinical isolates were determined by broth microdilution in four replicates in accordance with the European Committee on Antimicrobial Susceptibility Testing (EUCAST) and Clinical and Laboratory Standards Institute (CLSI) guidelines (45) with EUCAST clinical breakpoints used to determine antibiotic susceptibility and resistance (46). Phenotypic ESBL production was determined using combination disc diffusion testing with cefotaxime and ceftazidime with and without clavulanic acid according to CLSI standards (47). A ≥ 5 -mm increase in zone diameter for either antibiotic in combination with clavulanic acid compared to the zone diameter for the respective antibiotic alone, was determined to indicate phenotypic ESBL producers (47).

WGS, bioinformatic analysis, and SCK studies. WGS was performed using Illumina MiniSeq, High Output Reagent Cartridge (300 cycles) paired ends according to the manufacturer's instructions at the University of Queensland Centre for Clinical Research, Herston, Australia. Genomic analysis was conducted using a custom, in-house developed, microbial genomic analysis pipeline. WGS data were submitted to NCBI with BioProject accession no. [PRJNA762607](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA762607). Raw sequence read data was sent to the sequence read archive. SCK experiments were performed using an inoculum of $\sim 10^6$ CFU/mL over 48 h. The experimental design is presented in Table S1 in the supplemental material. The LLOQ was 2 log_{10} CFU/mL. Synergy was defined as a decrease in bacterial cells by $\geq 2\text{ log}_{10}$ between the antibiotics in combination and the most active antibiotic at 24 h (48). Detailed methods for WGS, bioinformatic analysis, and SCK are described in the supplemental material.

HFIM experiments. The HFIM experiments were conducted using a cellulosic cartridge (C3008; FiberCell Systems, Inc., Frederick, MD, USA) and performed over 7 days (49). Four ESBL-producing, piperacillin-tazobactam resistant *E. coli* isolates were tested in HFIM (Table 1). For each experiment, *E. coli* isolate was suspended in 24 mL of CAMHB and incubated at 37°C with constant agitation in a shaking water bath for a specific duration (according to growth curve assay) to attain a starting inoculum of $\sim 10^7$ CFU/mL. Bacterial samples (1 mL) were periodically withdrawn from the extra capillary space of the HFIM cartridge at 0, 2, 4, 6, 8, 10, 24, 28, 34, 48, 72, 96, 120, 144, and up to 168 h to perform bacterial quantification. Bacterial samples were washed twice to minimize drug carryover: centrifuged (3,500 g for 5 min) and resuspended in phosphate-buffered saline. An appropriately diluted 100 μ L of bacterial suspension was subsequently plated onto drug-free CAMHA and CAMHA supplemented with piperacillin-tazobactam (64 mg/L piperacillin/fixed 4 mg/L tazobactam) plus amikacin (32 mg/L), piperacillin-tazobactam (64 mg/L piperacillin/fixed 4 mg/L tazobactam) alone, amikacin (32 mg/L) alone, or meropenem (16 mg/L) alone. The LLOQ was 2 log₁₀ CFU/mL (i.e., counts less than 10 colonies per CAMHA plate were not considered). A reduction of >3 log₁₀ CFU/mL bacterial concentration from the baseline was considered indicative of bactericidal activity (49). The synergy is determined as ≥ 2 log₁₀ CFU/mL bacterial reduction when comparing antibiotics in combination with the most active antibiotic alone at the 24-h time point (50).

Piperacillin-tazobactam, amikacin, and meropenem clearances were simulated using peristaltic pumps (Masterflex L/S pump, USA). Drugs were directly infused into the central compartment using automated syringe pumps (New Era, model NE-1800). The free piperacillin-tazobactam, amikacin, and meropenem plasma exposures were simulated using population pharmacokinetic models (51–53), assuming patients with urosepsis, and a median creatinine clearance (CrCL) of 100 mL/min (41). Simulated clearance and half-life were 11 L/h and 1.10 h for piperacillin-tazobactam (51), 5.58 L/h and 4.20 h for amikacin (53), and 9.38 L/h and 2.22 h for meropenem (52). Simulated dosing regimens were piperacillin-tazobactam 4.5 g, 6-hourly, as monotherapy and in combination with amikacin 30 mg/kg of body weight, 24-hourly; amikacin 30 mg/kg of body weight, 24-hourly, as monotherapy, and meropenem 1 g, 8-hourly as monotherapy. All dosing regimens were administered as 30-min infusions and given over 7 days of experimental duration in the HFIM. A no-treatment growth control arm ran simultaneously in similar conditions as the treatment arm for each HFIM experiment. The experimental arms for each *E. coli* isolate were as follows: arm A-piperacillin-tazobactam and amikacin combination therapy; arm B-piperacillin-tazobactam monotherapy; arm C-amikacin monotherapy; arm D-meropenem monotherapy; arm E-no-treatment growth control. Samples (1 mL) were collected periodically from the outflow of the central reservoir in the HFIM at 0.5, 1, 2, 4, 5.8, 7.8, 10, 23.8, 24.5, 25, 28, 29.8, 31.8, 47.8, 48.5, 49, 52, 53.8, 55.8, 71.8, 72.5, 73, 76, 77.8, 79.8, 143.8, 144.5, 145, 148, 149.8, 151.8, and 168 h to measure the concentrations of studied drugs for pharmacokinetic analysis. Collected samples in cryovials were immediately stored at -80°C until further assay.

Piperacillin-tazobactam, amikacin, and meropenem assay. Determination of piperacillin and tazobactam concentrations in CAMHB were performed by a validated ultra-high performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) method on a Nexera liquid chromatograph connected to an 8030+ triple quadrupole mass spectrometer (Shimadzu, Kyoto, Japan). For piperacillin, the precision was within 3.3% and the accuracy was within 1.2% at 2.61, 21.8, 69.6, and 174 mg/L of piperacillin. For tazobactam, the precision was within 6.4% and the accuracy was within -3.3% at 0.326, 2.72, 8.70, and 21.8 mg/L of tazobactam.

Piperacillin was also detected in CAMHB using an ultra-high performance liquid chromatography-photo diode array (UHPLC-PDA) method on a Nexera2 liquid chromatograph connected to an SPD-M30 photo diode array detector (Shimadzu, Kyoto, Japan). Peaks for the analyte were detected at 242 nm. Two presumed β -lactamase-degradation products of piperacillin were detected, and concentrations were estimated. Amikacin in CAMHB was measured using a validated UHPLC-MS/MS method. The precision was within 9.9% and the accuracy was within -3.9% at amikacin concentrations of 1.5, 15, 40, and 80 (as quality controls (QC) dilution) mg/L. Meropenem in CAMHB was measured by an UHPLC-PDA method. The precision was within 3.8%, and the accuracy was within -6.4% at meropenem concentrations of 1.5, 25, and 80 mg/L. Test samples were analyzed in batches with calibrators, and quality controls and results were subject to pre-established batch acceptance criteria (US FDA 2018). (Detailed assay methods are provided in the supplemental material).

SUPPLEMENTAL MATERIAL

Supplemental material is available online only.

SUPPLEMENTAL FILE 1, PDF file, 0.5 MB.

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We declare no conflicts of interest.

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